

POS 6737: Political Data Analysis

University of Florida

Syllabus: Autumn 2025

Instructor:	Dr. Drew Rosenberg	Class location:	Weimer Hall 1098
Office:	210 Anderson Hall	Class time:	Tu, 08.30–11.30
Email:	andrewrosenberg@ufl.edu	Office hours:	W, 10.00–12.00, by appointment

TA:	Jain Choi	Recitation loc:	TBD
Office:	330 Anderson Hall	Recitation time:	TBD
Email:	jainchoi@ufl.edu	Office hours:	M, 10.00–12.00

Office Hours Links

Rosenberg: <https://calendly.com/asrosenberg>.
Choi: <https://calendly.com/choijain-jane>.

Course Description

Political Science 6737 is an introduction to probability and statistics targeted toward political science PhD students. A primary purpose of the course is to build a strong foundation for regression and generalized linear models, which will be studied in great depth in Political Science 6747 and beyond. To accomplish this goal, we will study the basics of probability theory, properties of random variables, asymptotic approximations, methods for developing and evaluating statistical estimators, and hypothesis testing. In addition, the course will provide a hands-on introduction to statistical computing.

The course will be taught as a combination of lectures by the instructor and practical exercises at the computer.

Course Goals:

1. Students will gain experience using and understanding the basic foundations of all quantitative social science. This experience will allow them to conduct their own projects, progress to more advanced courses on regression and causal inference, and evaluate published work.

Expected Learning Outcomes:

- Learn basic probability axioms and rules and the moments of discrete and continuous random variables as well as be familiar with common named discrete and continuous random variables.
 - Use summary statistics and graphs to conduct exploratory data analysis.
 - Calculate probabilities, and derive the marginal and conditional distributions of bivariate random variables.
 - Conduct hypothesis tests.
 - Understand the fundamental problem of causal inference.
2. Students will develop modern statistical computing skills that will allow them to conduct data analysis at the PhD level.

Expected Learning Outcomes:

- Read data into R and do basic data cleaning.
- Learn good coding etiquette.
- Download packages, write functions, and debug code.
- Run a basic simulation study to solve probability exercises.
- Run and interpret simple regressions.
- Use R Markdown to present research findings in a beautiful format.

Course Materials:

Textbooks

We will use two textbooks in this course, one traditional and one less-traditional. The first book is:

Sean Gailmard. *Statistical Modeling and Inference for Social Science*. New York: Cambridge University Press, 2014.

We will also use the online interactive learning platform, **zyBooks**, for this course. zyBooks offers a unique and engaging way to study quantitative methods through a blend of concise text, interactive learning questions, animations, and embedded tools that help reinforce key concepts. Additionally, zyBooks includes resources and exercises specifically designed to help you learn R, a statistical programming language that will be essential for this course.

How to Access:

1. **Visit:** learn.zybooks.com
2. **Sign Up:** Click on 'Sign In' and create an account using your university email.

3. **Enter zyBook Code:** Upon signing in, you will be prompted to enter a zyBook code specific to this course. Our code is **UFLPOS6737RosenbergFall2025**.
4. **Subscribe:** Follow the on-screen instructions to subscribe to the zyBook for this course.

Why zyBooks?

Interactive Learning: Instead of passively reading through content, zyBooks encourages active participation through interactive elements that test your understanding as you progress.

Immediate Feedback: The platform offers instant feedback on activities, allowing you to learn from mistakes and understand concepts more deeply.

Engagement: The multimedia approach of zyBooks—combining text, animations, and interactive exercises—makes the learning process more engaging and effective.

Learning R: zyBooks provides targeted resources and exercises to help you learn statistical computing in R.

I also create very detailed lecture slides and make them available each week. You should think about these slides and the lectures as part of the weekly course readings. This multi-book approach might seem weird to some of you, but there are thousands of possible books for this course, and you probably won't like most of them. If you are someone who learns best from reading textbooks and are having difficulty with the slides, then please come talk to me and I will find you a reading that will suit the topic.

We may also use selections from another more advanced textbook, but I will provide them to you when necessary. This book tends to present some ideas better than the Gailmard book.

Richard J. Larsen and Morris L. Marx. *An Introduction to Mathematical Statistics and its Applications*. Englewood Cliffs: Prentice-Hall, 2011.

Third, this is a useful guide to using **R** that we will use. I will assign modules from this guide:

[fasteR: Fast Lane to Learning R](#).

Finally, this is one of the best books available on **R** programming, and I would use it if I taught a semester-long data science course.

Jared P. Lander. *R for Everyone: Advanced Analytics and Graphics*. 2nd ed. Addison-Wesley, 2017.

There are countless introductory textbooks on probability and statistics and R. You may wish to consult any of these for this class, but they are by no means required:

George Casella and Roger L. Berger. *Statistical Inference*. Pacific Grove: Duxbury Press, 2002.

Morris H. DeGroot and Mark J. Schervish. *Probability and Statistics*. New York: Addison-Wesley, 2012.

Paul M. Kellstedt and Guy D. Whitten. *The Fundamentals of Political Science Research*. Cambridge University Press, 2018.

Hadley Wickham and Garrett Grolemund. *R for Data Science: Import, Tidy, Transform, Visualize, and Model Data*. Sebastopol: O'Reilly Media, Inc., 2016 (FREE at <https://r4ds.had.co.nz/>).

Statistical Software

NB: This is the section that will seem overwhelming at first. Please bear with me; I promise that the computer skills you learn now will pay massive dividends later.

Proficiency in political data analysis requires one to analyze political data! With this end in mind, we will be playing with data from the very first week of this course. We will use the open source and free statistical software **R** in our course: <http://www.r-project.org/>.

What is R and why use it?

- Widely-used in academia and industries
- Open-source and free
- Power and flexibility
- Graphical capabilities
- Learning R = learning basic programming
- When you accomplish things, it will feel awesome
- And more!

I recommend that you also install the free RStudio interface (<http://www.rstudio.com/>), which makes working with **R** a little easier. The first problem set will walk you through the process of installing **R** and RStudio on your own computer/laptop. There are plenty of free resources for **R** to be found on the internet. Google will get you very far in many instances.

R Markdown

Markdown is a basic mark-up language that allows you to use simple syntax to make beautiful documents. R Markdown is a special version of Markdown that allows you to embed data analysis into text documents. This might not seem cool, but it is: it allows you to create reproducible analyses that anyone can replicate. When you use R Markdown in RStudio, you can output PDF, HTML, and Microsoft Word documents. It's pretty cool. Here is an

introduction: <https://rmarkdown.rstudio.com/lesson-1.html>. We will also spend the first problem set getting you up and running!

L^AT_EX

Although I encourage you to use R Markdown, you can also familiarize yourself with the L^AT_EX typesetting environment. L^AT_EX documents look a bit like scary computer code, but they allow you to make beautiful documents with perfect citations. However, R Markdown actually uses L^AT_EX “under the hood,” so they are equivalent. In addition, <https://www.overleaf.com/> is an online L^AT_EX editor that makes typesetting and collaboration quite easy. Please come talk to me more if you need help.

Overleaf’s introduction is very good and there are many other resources online:

[https://www.overleaf.com/learn/latex/Free_online_introduction_to_LaTeX_\(part_1\)](https://www.overleaf.com/learn/latex/Free_online_introduction_to_LaTeX_(part_1))

Assignments:

- **ZYBOOKS PARTICIPATION AND CHALLENGE ACTIVITIES (5% and 20%, respectfully):** Throughout the course, you will be required to complete participation and challenge activities on the zyBooks platform. These activities are designed to reinforce the material covered in lectures and readings. Participation activities include interactive exercises and quizzes that allow you to engage with the content in an active way. **Participation activities are graded based on completion only.** Challenge activities are more in-depth tasks that test your understanding of the concepts and require you to apply what you’ve learned in practical scenarios. **Challenge activities are graded based on completion and correctness.** Regular engagement with zyBooks will help solidify your knowledge and prepare you for the problem sets and exams.
- **HANDS-ON PROBLEM SETS (25%):** There will be additional problem sets almost every week. The problem sets typically consist of a set of theoretical and conceptual questions and a hands-on data analysis portion. The purpose of the problem sets is to give you practice doing data analysis and engaging with each week’s concepts. Students are encouraged to work on the problem sets in small groups, i.e. you should discuss possible answers with your fellow students. It is good practice to first try to develop answers on your own and then meet in a group setting to discuss potential difficulties. While group discussion and work is explicitly encouraged, you are required to write and hand in your own computer code and final write-up of the answers. **DO NOT** simply copy computer code or answers from your classmates. Write-ups have to be provided in a well-formatted, electronic format (e.g. R Markdown). I will not accept any late homework assignments. To accommodate your busy schedules, **I will drop your two lowest homework scores when calculating your final grades.** These will be pretty hard, so don’t worry. See below for more reassurance.
- **PROBLEM SET GRADING PROCEDURE**

- ✓+ (40 points; 100%) Problem set is 100% complete. Every question was attempted and answered, and all are correct. Document is clean and easy to follow. Code is well-written. Work is exceptional. *These are rare.*
- ✓ (37 points; 93%) Problem set is 75—99% complete and most answers are correct. *This is the expected level of performance.*
- ✓− (25 points; ~ 63%) Problem set is less than 75% complete and/or most answers are incorrect. This indicates that you need to improve next *and make an appointment to come talk to me.* This is not an indictment of your ability to do well in this course!
- WEEKLY CHECK-INS (5%): After class each week you will be responsible for letting me know what was confusing from lecture and/or what you wanted to hear more about. You will submit this feedback on my website: <https://www.asrosenberg.com/political-data-analysis>.
- MIDTERM EXAM (15%): There will be an in-class, closed-note midterm that covers the conceptual material from the first half of the course.
- FINAL EXAM (DECEMBER 9) (30%): There will be an in-class, closed-note final exam that covers the conceptual material from the entire course.
- ATTENDANCE POLICY: We will meet once a week during the semester. You can expect me to be prepared, give the lecture and answer questions. When you come to class, please also be prepared. I will not require attendance, but class is a resource to *you*. The classroom is a great place to exchange ideas, meet your classmates, and ask questions. Regular attendance is also encouraged because lectures and practical sessions are tightly linked to weekly assignments, the midterm and final. If you do not attend regularly, it will be difficult to pass the class. Requirements for class attendance and make-up exams, assignments, and other work in this course are consistent with university policies that can be found at: <https://catalog.ufl.edu/ugrad/current/regulations/info/attendance.aspx>.
- SUMMARY OF MOST IMPORTANT DATES:
 - OCT 21: Midterm
 - DEC 9: Final Exam
- GRADING SCHEDULE:

	A	93–100%	A−	90–92%
B+	87–89%	B	84–86%	B−
C+	77–79%	C	74–76%	C−
D+	67–69%	D	64–66%	D−
E	< 60%			

How to Succeed in this Class

Statistics classes are really hard. Most people find this course challenging and we cover a lot of stuff. But you can do it; you have nothing to fear! All successful people struggle with this class. It's not just you.

Your responsibility is to work hard, do your best, and communicate with me. You cannot learn this stuff if you don't put in the time. I can't help you if you don't turn in assignments. I can't help you if I don't know there is a problem. Here are some more specific resources.

1. Lectures. I will lecture during most of our time together. I will speak over the slides and often provide off-the-cuff examples and explanations. The goal is to help you understand the material, so please let me know in class how I can help. Participate! Ask me stuff!
2. Readings. You should do the reading before you come to class. This includes both the zyBooks participation activities and the Gailmard selections.
3. Weekly Check-ins. See above. I can't help you unless you let me know what you need help with.
4. Lecture slides. The slides will be available on Canvas after class. You have to read them and ask questions about them if there are things you don't understand.
5. Example Code. I will provide example code and in-class exercises to help you learn R. Please come to class prepared to participate.
6. Recitation. Our TA will host an optional recitation to help reinforce the weekly concepts. You should go to them.
7. Office hours. Don't be afraid to come talk to me about broader conceptual issues and specific things you don't understand.
8. The Internet. There are infinite free resources online that will help with the conceptual and computing aspects of the course.
9. Your classmates. You will learn more from each other than from me. Form a study group!

If you have any questions or can think of anything else that would be useful for you, then please come talk to me. To reiterate: if you work hard and put in the time, then I can provide help that meets your needs.

Policies and procedures

Communication and logistics: Email

Please email me with any pressing questions or concerns. However, do not expect immediate replies. I often do not check my email on the weekends or in the evening.

Office Hours

I hold three hours of office hours per week, but you may arrange a meeting outside of those hours if you are unavailable during this time. Please make use of office hours, as that is the time I allocate to be 100% available to you. If you have any questions or are having difficulty completing course requirements, please come see me as soon as possible. *Use the Calendly link at the top of this syllabus and on my website to book a meeting.*

Computers

I will provide example computer code and in-class exercises to help you learn R. I encourage you to bring your computer to class each week in order to participate. Please reach out to me as soon as possible if you have trouble downloading R on your computer.

Collaboration Policy

I encourage students to work together on the problem sets, but you must write your own solutions (this includes code). However, I *strongly* suggest that you try all the problems before consulting others. The midterm will be very difficult if you have no experience working on your own.

AI Policy

Do not use any AI tools in this course. AI tools are good at coding, but you need to learn this stuff on your own.

The UF student conduct handbook states, “A Student must not submit as their own work any academic work in any form that the Student purchased or otherwise obtained from an outside source, including but not limited to: academic work in any form generated by an Entity; academic materials in any form prepared by a commercial or individual vendor of academic materials; a collection of research papers, tests, or academic materials maintained by a Student Organization or other entity or person, or any other sources of academic work.”

Entity “include[s] but is not limited to generative artificial intelligence, large language models, content generation bots, or other non-human intelligence or digital tools.”

Assignment dispensation policy

If a student is unable to complete an assignment, they will be allowed to turn it in late only if the absence is due to a *documented* medical, family, or similar serious emergency, observance of religious holy days (which requires written notification to the instructor at least 14 days prior to the due date), or properly documented University-sponsored planned activities. *Incomplete assignments or exams in all other cases will result in a score of zero.* If you become aware that you will not be able to complete an assignment or final project ahead of time, please contact the instructor and seek permission for an extension as soon as possible.

This course complies with all UF academic policies. For information on those policies and for resources for students, please see this [link](#).

Online Course Evaluations

Students are expected to provide professional and respectful feedback on the quality of instruction in this course by completing course evaluations online. Students can complete evaluations in three ways:

1. The email they receive from GatorEvals,
2. Their Canvas course menu under GatorEvals, or
3. The central portal at <https://my-ufl.bluer.com>

Guidance on how to provide constructive feedback is available at <https://gatorevals.ua.ufl.edu/students/>. Students will be notified when the evaluation period opens. Summaries of course evaluation results are available to students at <https://gatorevals.ua.ufl.edu/public-results/>.

Course Overview and Schedule:

Week 1 (Aug 26): Introduction and Descriptive Statistics I

Introduction to the class, general requirements, and logistics.

- Readings:
 - Skim Gailmard, Ch. 1.
 - Lesson 1 of fastR.
- Assignments:
 - zyBooks PA (8/29)
 - zyBooks CA (8/29)
 - PS-1

Week 2 (Sept 2): Descriptive Statistics II

- Readings:
 - Gailmard, Ch. 2, through pp. 56.
 - Lessons 2–3 of fastR.
- Assignments:
 - zyBooks PA (9/2)
 - zyBooks CA (9/5)

- PS-2

Week 3 (Sept 9): Probability Theory

- Readings:
 - Gailmard, Ch. 3–4.2.
 - Lessons 4–7 of fastR.
- Assignments:
 - zyBooks PA (9/9)
 - zyBooks CA (9/12)
 - PS-3

Week 4 (Sep 16): Discrete and Continuous Random Variables; Joint/Marginal/Conditional Distributions

- Readings:
 - Gailmard, Ch. 4.3–4.7.
 - lessons 8–13 of fastR.
- Assignments:
 - zyBooks PA (9/16)
 - PS-4

Week 5 (Sep 30): No Class

Week 6 (Oct 7): Expectation and Variance

- Readings:
 - lessons 14–15 of fastR.
 - Gailmard, 5.1.1–5.1.3 and 5.2.1–5.2.4 (skip “Important Properties of the DGP Regression” sub-section).
- Assignments:
 - PS-5

Week 7 (Oct 14): Probability and Models

- Readings:
 - Gailmard, Ch. 6 (difficult, but don’t despair).

- Lesson 16 of fastR.
- Assignments:
 - zyBooks PA (10/14)
 - zyBooks CA (10/17)
 - PS-6

Week 8 (Oct 21): MIDTERM

This speaks for itself! It's a closed-book exam and everything will be ok!

Week 9 (Oct 21): Normal Distribution; Sampling Distributions

- Readings:
 - Gailmard, Ch. 7.
 - Lesson 17 of fastR.
- Assignments:
 - zyBooks PA (10/21)
 - zyBooks CA (10/24)
 - PS-7

Week 10 (Oct 28): Hypothesis Testing I

- Readings:
 - Gailmard, Ch. 8.1–8.3.
 - Lesson 18 of fastR.
- Assignments:
 - zyBooks PA (10/28)
 - zyBooks CA (10/31)
 - PS-8

Week 11 (Nov 4): Hypothesis Testing II

- Readings:
 - Gailmard, Ch. 8.4–8.8.
 - Lesson 20 of fastR.

- Assignments:
 - zyBooks PA (11/4)
 - PS-9

Week 12 (Nov 18): Interval Estimation

- Readings:
 - Gailmard, Ch. 9.1.
 - Lesson 21 of fastR.
- Assignments:
 - zyBooks PA (11/18)
 - zyBooks CA (11/21)
 - PS-10

Week 13 (Dec 2): Point Estimation

- Readings:
 - Gailmard, Ch. 9.2–9.3.
 - Lesson 24 of fastR.
- Assignments:
 - PS-11